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Ph: 91-11-26384723, Fax: 91-11-26384722
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of embedded systems, and the integration of embedded systems with more complex applications.

One of the obvious examples in today's scenario is the mobile phone. A mobile phone is almost a personal computer today—it holds critical applications as well as data. It connects to a variety of networks to transmit and receive data. A security breach could prove to be very expensive.

Likewise, consider other embedded systems that are used in defence equipment or industrial machinery. These are very likely to connect to a larger army network or enterprise-wide application. A security breach in one little sensor could prove to be fatal to the whole network. Thanks to these scary scenarios, the demand for embedded platforms with tight security features is very high today.

This, however, is no mean task—because once again, the rules of embedded systems design come into play. The security features have to be imbibed into the system without increasing the size or decreasing the speed of the system noticeably.

5 The start of a cloud computing era

"Last year witnessed many trends in the embedded space, like OS vendors also developing processors and vice versa, the emergence of ARM as a powerful competitor to Intel chips, the consolidation of semiconductor vendors and multi-core support in embedded systems (say, ARM Cortex-A9), but the greatest was the foundation of cloud computing laid in the embedded systems," says Sridharan Mani, director and CEO, American Megatrends India.

The last year has seen the IT industry working seriously on combining the benefits of embedded computing and cloud computing—metaphorically, taking the little drops of water to the mighty ocean. Scalable edge node (SEN) and server platforms happen to be the key step in this transition. SEN is a dense, flexible hardware platform that would reside on the edge of a net-

work of embedded systems—be it in a factory floor or an aeroplane.

The main job of SEN is to provide connectivity between the various applications of the network and the cloud, so that the intelligence gained by the embedded systems can be transferred to a larger application on the cloud for analysis and processing. It is estimated that in such a scenario, over 80 per cent of the analysis and processing will be shifted from the embedded systems to the cloud.

"SENs would enable clouds to be deployed across different value propositions and also enable scalability. The transition has started to happen and it would be interesting to see how it scales," says Mani.

Currently, embedded cloud computing solutions are available from companies like Digi, Eurotech and National Instruments. They have been deployed in smart power grids, medical communicators, environment monitors and defence equipment. Microsoft also recently announced that it is in a position to offer customers a composite platform that spans from the endpoint device to the back-end cloud computing infrastructure. Much more of such work is happening across the industry, and it would be interesting to wait and watch what the union of embedded and cloud computing is going to yield.

With such high capabilities, it is not surprising that embedded systems are finding more and more applications—in the dashboards of automobiles, entertainment devices, mobile phones, inventory systems, industrial equipment, security and surveillance networks, medical equipment and so on. This is a good vicious cycle—with technological development encouraging more usage, and increasing usage fuelling more technological growth. We are surely in for an exciting time with these technological trends strengthening and many more emerging in the near future. ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai